ServiceMate – Database Schema Design Document

1. Project Title

ServiceMate – Local Service Booking Platform

2. Overview

ServiceMate connects customers with nearby service providers such as electricians, plumbers, carpenters, and cleaners. The platform enables users to discover services, book appointments, make payments, and provide reviews. The database schema manages users, providers, services, bookings, payments, and reviews using relational database principles.

3. Database Technology

Database System: MySQL / PostgreSQL

Database Design Type: Relational Database

Schema Model: Entity-Relationship Model (ER Model)

4. Database Entities

Users – Stores platform users

Service_Providers – Stores provider details

Services – Stores services offered

Bookings – Stores booking information

Reviews – Stores ratings and feedback

Payments – Stores payment transaction details

5. Table Structures

Users Table

Column	Data Type	Description
user_id	INT (PK)	Unique user identifier
name	VARCHAR	User full name
email	VARCHAR	User email (unique)
password	VARCHAR	Encrypted password
phone	VARCHAR	Contact number
role	ENUM	customer/provider/admin
created_at	TIMESTAMP	Account creation time

Service_Providers Table

Column	Data Type	Description
provider_id	INT (PK)	Provider ID
user_id	INT (FK)	Reference to Users table
service_category	VARCHAR	Service type
experience	INT	Years of experience
location	VARCHAR	Provider location
availability	BOOLEAN	Availability status

Services Table

Column	Data Type	Description
service_id	INT (PK)	Service ID
provider_id	INT (FK)	Provider reference
service_name	VARCHAR	Name of service
description	TEXT	Service details
price	DECIMAL	Service cost

Bookings Table

Column	Data Type	Description
booking_id	INT (PK)	Booking ID
user_id	INT (FK)	Customer reference
service_id	INT (FK)	Service reference
booking_date	DATE	Booking date
status	VARCHAR	Booking status

Reviews Table

Column	Data Type	Description
review_id	INT (PK)	Review ID
booking_id	INT (FK)	Booking reference
rating	INT	Rating 1–5
comment	TEXT	Customer feedback
created_at	TIMESTAMP	Review time

Payments Table

Column	Data Type	Description
payment_id	INT (PK)	Payment ID
booking_id	INT (FK)	Booking reference
amount	DECIMAL	Payment amount
payment_status	VARCHAR	Payment status
payment_method	VARCHAR	Payment method
payment_date	TIMESTAMP	Payment time

6. Key Relationships

Users → Service_Providers (One user can be a provider)

Service_Providers → Services (One provider can offer many services)

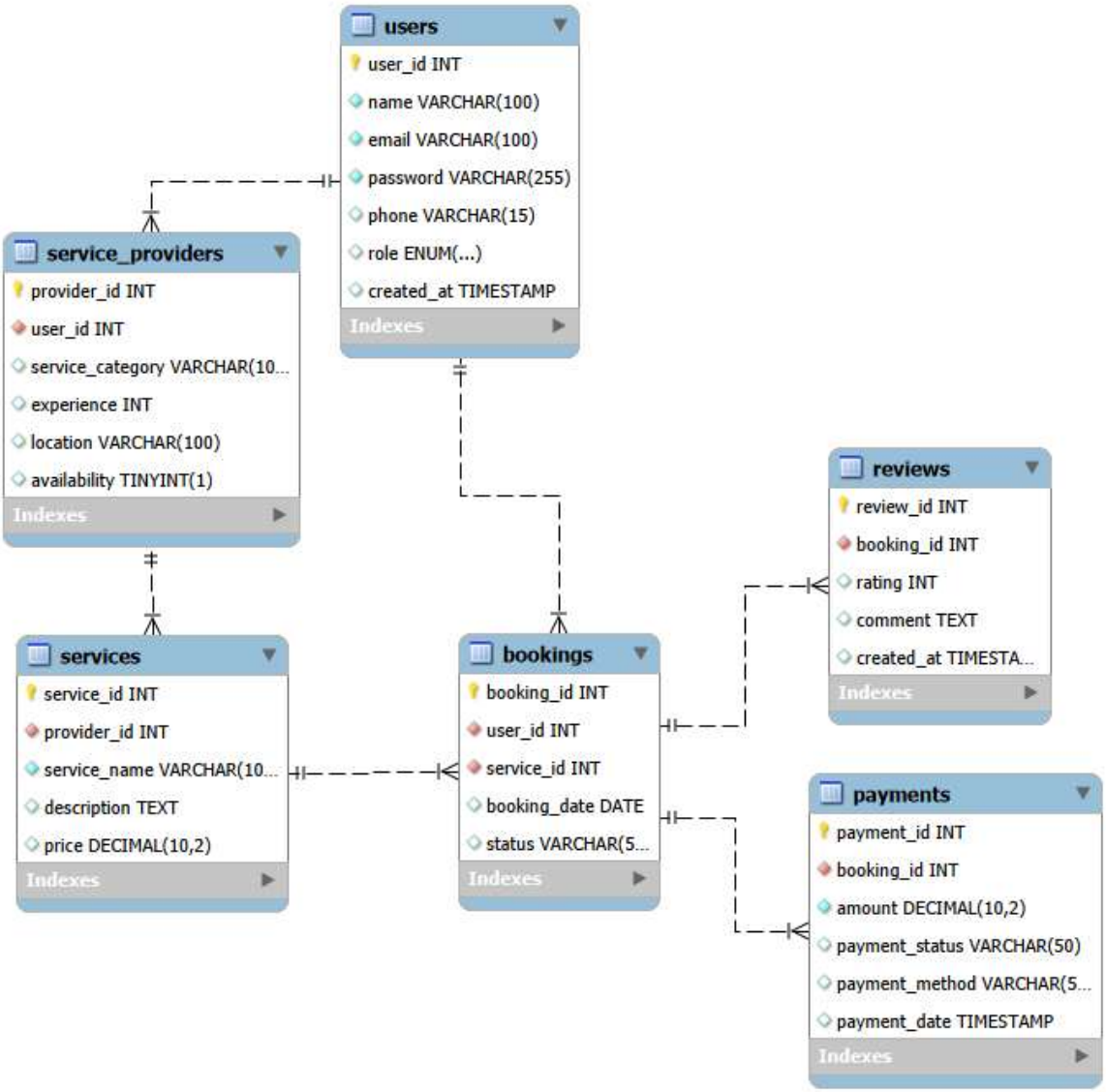
Users → Bookings (Customers create bookings)

Services → Bookings (Services can be booked multiple times)

Bookings → Reviews (Booking can have a review)

Bookings → Payments (Each booking has a payment record)

7. ER Diagram



8. Database Implementation

Database Creation

CREATE DATABASE IF NOT EXISTS servicemate;

USE servicemate;

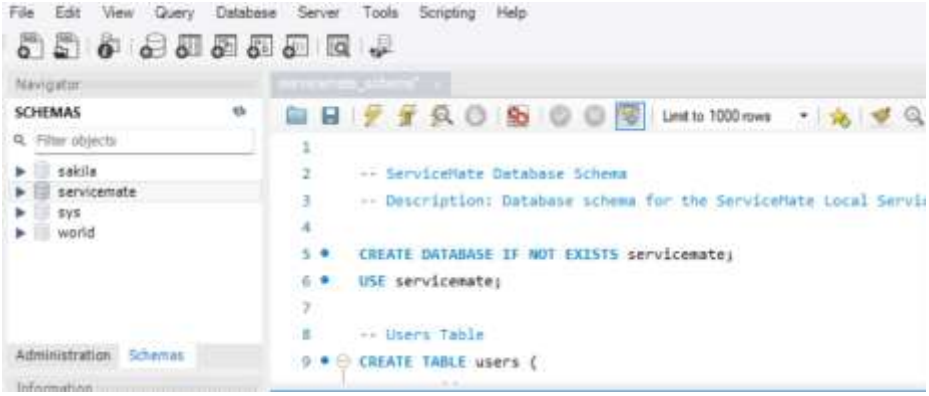


Table Created

CREATE TABLE users (

user_id INT AUTO_INCREMENT PRIMARY KEY,

name VARCHAR(100) NOT NULL,

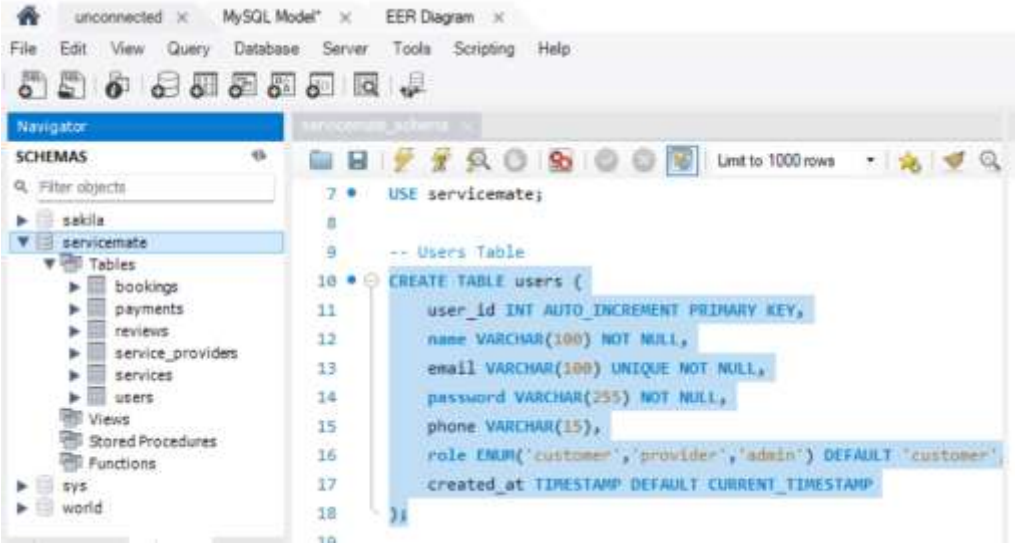
email VARCHAR(100) UNIQUE NOT NULL,

password VARCHAR(255) NOT NULL,

phone VARCHAR(15),

role ENUM('customer','provider','admin') DEFAULT 'customer',

created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP);



9. Sample Data Insertion

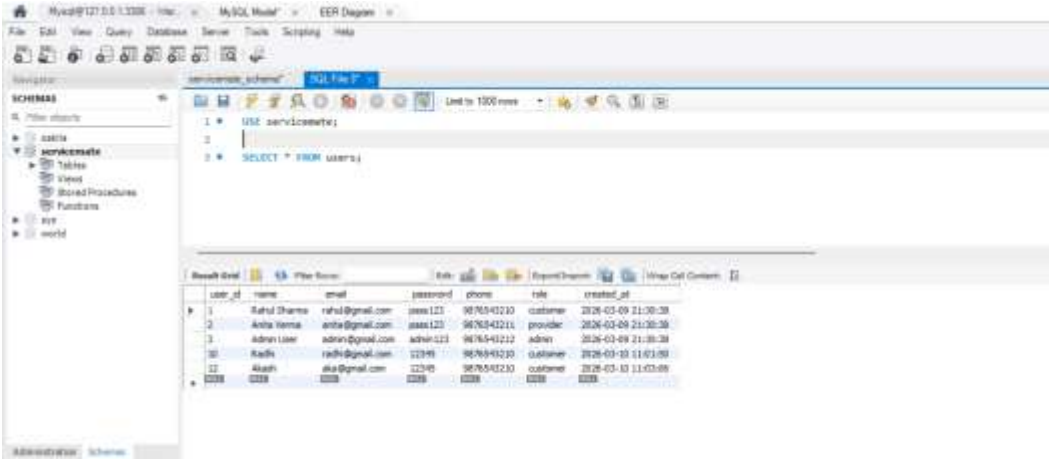
User Table

USE servicemate;

INSERT INTO users (name,email,password,phone,role)

VALUES ('Rahul','rahul@gmail.com','12345','9876543210','Customer');

SELECT * FROM users;

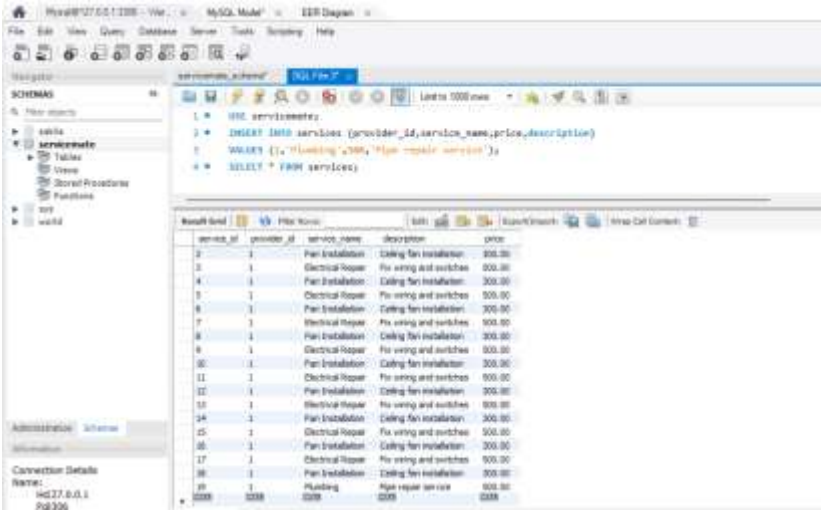


Service Table

USE servicemate;

INSERT INTO services (provider_id,service_name,price,description)

VALUES (1,'Plumbing',500,'Pipe repair service'); SELECT * FROM services;



BookingTable

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USE servicemate;

INSERT INTO bookings (user_id,service_id,booking_date,status)

VALUES (1,1,'2026-03-10','Pending');

SELECT * FROM bookings;
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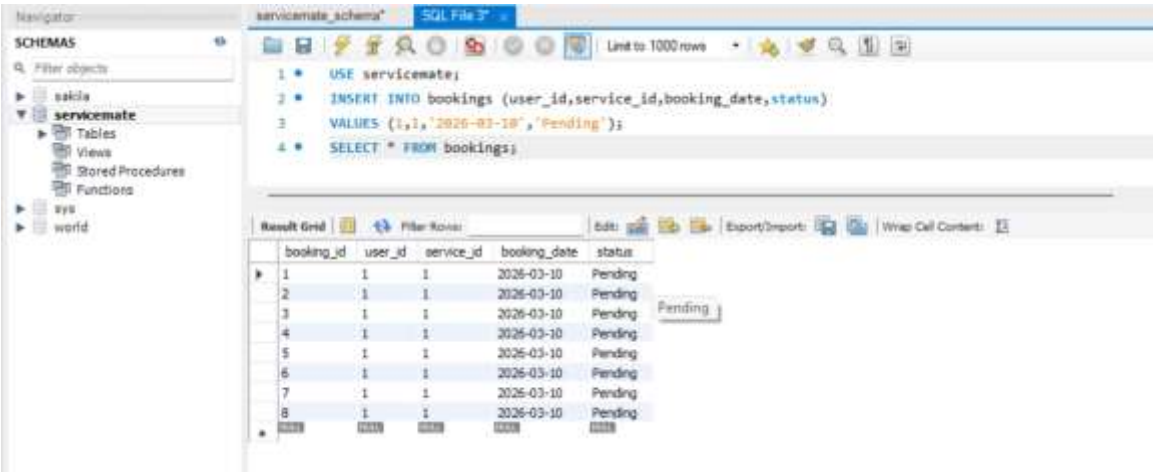

10. Future Improvements

Location-based service search, real-time provider availability, payment gateway integration, notification system, and mobile app integration.